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CRYPTOCURRENCY VOLATILITY AND ITS SPILLOVER EFFECTS ON TRADITIONAL FINANCIAL MARKETS

Bachelor's thesis

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and all works, important standpoints and data by other authors
have been properly referenced and the same paper
has not been previously presented for grading.

The document length is 7404 words from the introduction to the end of the conclusion.

Nurmukhammed Taiynbas ……

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TABLE OF CONTENTS

ABSTRACT 4
INTRODUCTION 5
1. LITERATURE REVIEW 7
1.1. Introduction to Cryptocurrency 7
1.2. Theoretical background 9
1.2.1. Volatility in the Cryptocurrency Market 9
1.2.2. Impact on Traditional Financial Markets 10
1.3. Empirical study evidence 12
2. DATA AND METHODOLOGY 14
2.1. Data 14
2.2. Methodology 16
2.3. Correlation matrix 19
3. VOLATILITY SPILLOVER ANALYSIS 21
3.1. DCC-GARCH model estimation 21
3.2. Dynamic Conditional Correlations 23
3.2.1. Stock markets analysis 23
3.2.2. Commodities market analysis 26
3.3. Discussion 28
CONCLUSION 30
LIST OF REFERENCES 32
APPENDICES 42
Appendix 1. Link to R-code and data 42
Appendix 2. Robustness check: Results of ADF test 43
Appendix 3. Robustness check: Results of ARCH-LM test before model estimation 44
Appendix 4. Robustness check: Results of Ljung-Box test 45
Appendix 5. Robustness check: Results of ARCH-LM test after model estimation 46
Appendix 6. Robustness check: Results of Engle-Ng test 47
Appendix 7. Non-exclusive licence 48

ABSTRACT

This paper studies the volatility spillover effects from the cryptocurrency market to traditional financial markets. This paper's main research problem revolves around whether cryptocurrency volatility significantly impacts the volatility of traditional financial markets represented by equity and commodity markets. Additionally, this work examines whether these effects differ between countries with high and low levels of cryptocurrency adoption. To explore this research problem, the study applies Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroscedasticity (DCC-GARCH) and interprets the results of daily dynamic correlation. Return data of BITW cryptocurrency index, major stock indices, gold, and WTI crude oil were selected to conduct the analysis.

The main conclusion of this study is that cryptocurrencies and traditional financial markets operate independently under normal market conditions. However, during periods of positive or negative market shocks, the interdependence might become more intense. Additionally, the author found no evidence that the level of cryptocurrency adoption in a country influences the overall pattern of volatility spillover effects.

Keywords: cryptocurrency volatility, volatility spillovers, DCC-GARCH model, traditional financial markets

INTRODUCTION

The rapid growth of the cryptocurrency market has become a notable challenge to the global financial system. Since the introduction of Bitcoin in 2008, crypto markets have transformed from a niche digital asset class to an integral part of the financial world. A key feature of this market is its persistent volatility and speculative behaviour, which is not just a simple trademark, but a factor that poses a potential contagion risk to financial markets globally, especially through volatility spillovers. Therefore, this is a crucial area to question and analyse how exactly those effects might impact traditional financial markets.

The objective of this thesis is to investigate the extent to which volatility in the cryptocurrency market affects traditional financial markets, with a specific focus on stock indices across countries with varying levels of cryptocurrency adoption. Many studies miss the role of cryptocurrency adoption levels across different markets. This is an important point to consider since the level of crypto adoption within the country can directly impact the sensitivity to volatility due to higher or lower integration of the crypto market. For that particular reason, this study aims to determine if the volatility spillover from cryptocurrencies to traditional financial assets is influenced by the level of cryptocurrency adoption. Additionally, it examines the relationship between cryptocurrency market volatility and commodities markets. Specifically, this study seeks to answer the following research questions:

Does volatility in the cryptocurrency market, have a significant impact on the volatility of stock prices in countries with high versus low cryptocurrency adoption rates?
Does volatility in the cryptocurrency market, have a significant impact on the volatility of gold prices?
Does volatility in the cryptocurrency market have a significant impact on the volatility of oil prices?

Based on similar studies that also examined a similar topic, this paper employs the most popular approach of Dynamic Conditional Correlation Generalized Autoregressive Conditional

Heteroscedasticity (DCC-GARCH) to conduct an empirical study. The model is estimated based on daily price data for the BITW index, gold, WTI crude oil and stock indices of six selected countries: India, Brazil, the UK, Germany, South Africa and Japan from December 2020 to November 2024.

This paper is organized as follows. Following the introduction, Chapter 1 provides a full literature review, starting from a historical review of cryptocurrency development, a theoretical framework on crypto volatility and its propagation channels and prior empirical studies. Chapter 2 presents the data and methodology used in the study, justifying the choice of the dataset used for model estimation and further explanation of the chosen model. This chapter describes the data transformation process, the model specification, the conditions required for proper model estimation, and the preliminary analysis of static correlation. Chapter 3 presents the estimation results of the DCC-GARCH model, followed by a discussion part interpreting the results with factors that may have affected market dynamics, results of model diagnostics, and limitations of the study. The final chapter presents a final summary conclusion.

1. LITERATURE REVIEW

1.1. Introduction to Cryptocurrency

The initial idea of cryptocurrency was proposed by American cryptographer David Chaum (Vejačka, 2014; Rice, 2019; Zakarneh et al., 2022) who created one of the earliest versions of the digital currency "Ecash," which shared some similarities with Bitcoin. Although Chaum's ideas date back to the 1980s, significant attention to cryptocurrency emerged only in 2008 when an anonymous individual or group named Satoshi Nakamoto introduced Bitcoin as a "peer-to-peer distributed timestamp server" which was mainly described as a "chain of digital signatures" (Nakamoto, 2008). The unique feature behind Bitcoin was its decentralized nature and according to Farell (2015), Nakamoto's base idea was to introduce a currency that would eliminate trusted central authority between transactions.

The decentralized system had additional advantages such as low transaction fees and anonymity (Shah et al., 2023), making it attractive both as an alternative and as a challenge to traditional financial institutions (Bibi, 2023). Cryptocurrency systems eliminate the need for centralized authority by implementing blockchain technology, which is explained as 'distributed consensus' among network participants that provides secure and transparent distribution of transactions (Narayanan, 2016). One more distinct feature of Bitcoin from fiat currencies is its finite supply (21 million) which contributes to its value, making it an attractive investment option (Hossaion et al., 2023) and according to Bjerg (2016), scarcity and the mining process are argued to contribute to its perceived value, aligning with the principles of commodity theory. Given all these characteristics, Bitcoin has evolved from a simple digital "concept" into an asset class that attracts a wide range of investors. Nevertheless, historically Bitcoin did not achieve immediate success in the market and Figure 1 shows that Bitcoin price fluctuated around 10-15 USD from late 2011 until January 2013. Still, after that period, the price sharply increased reaching a peak of 250 USD by the end of the first quarter in 2013, with a further abnormal increase to 1200 USD by 2024.

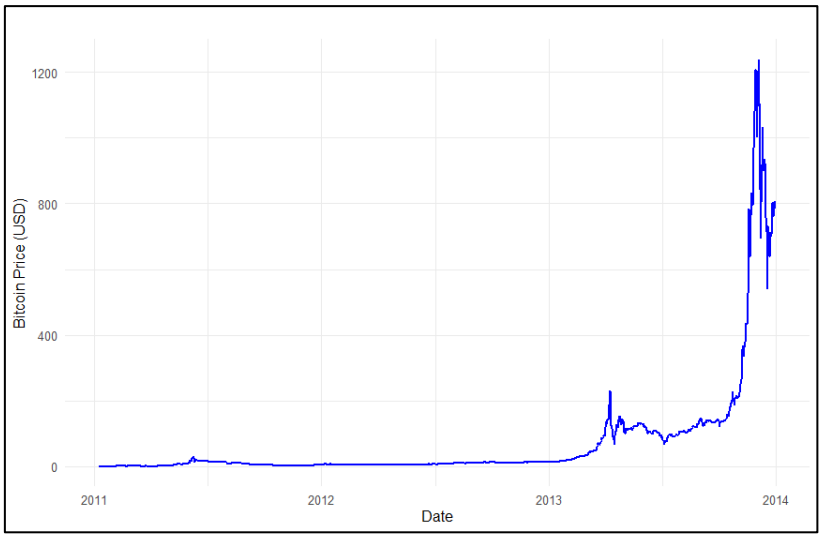

Figure 1. Bitcoin price from January 2011 to December 2013
Source: Author's calculations, see Appendix 1

Since Bitcoin's inception, numerous alternative cryptocurrencies, known as altcoins, have entered the market, capturing investor's attention, while Bitcoin's market capitalization grew significantly, altcoins collectively grew even faster, reducing Bitcoin's market share from 91.84% (April 30, 2013) to 77.02% (December 30, 2014), from 89.66% to 42.42% in 2017 and by the end of 2022 this number went even lower with decreasing market share of Bitcoin to 39.98% (MacroMicro, 2024).

Altcoins have introduced some new features and improvements in speed, robustness, and privacy for the cryptocurrency market (Dziembowski, 2015) and according to Jalal et al. (2021) there are 2 main categories of altcoins, the first group follows the protocol as Bitcoin (Litecoin) and the second group which is built on entirely independent protocols with unique distributed consensus ledgers (Ethereum, Ripple). An increasing number of different cryptocurrencies are introduced daily. Some of them are not just simple alternatives to Bitcoin but provide unique ledger systems and are gaining an increasingly strong position within the market, which means that cryptocurrency evolved from a niche digital concept to a significant component of the financial landscape, with over 9,900 cryptocurrencies introduced as of November 2024 (Statista, 2024).

1.2. Theoretical background

Although the cryptocurrency market is still developing, research suggests it can extend beyond speculative trading to influence various socio-economic aspects (Ben Dhaou & Rohman, 2018; Abramova, 2018; Pantielieieva et al., 2019). For instance, this market introduces the potential to address social and environmental challenges, despite ongoing concerns about trust and cybersecurity (Shin & Rice, 2022). The impact of crypto assets already reaches beyond finance, affecting areas such as politics by providing alternative fundraising opportunities for marginalised groups (Golumbia, 2015), as well as education, innovation, and culture by redefining value and ownership concepts, particularly through Non-Fungible Tokens (NFTs) (Al-Saqaf & Seidler, 2017; Chalmers et al., 2022). However, it is equally important to acknowledge the problems cryptocurrency markets can create. For example, cryptocurrency mining raises environmental concerns due to its high electricity consumption (Chamanara et al., 2023), and the pseudonymous transaction system facilitates illegal activities, such as money laundering and other financial crimes (Panda & Jani, 2019).

1.2.1. Volatility in the Cryptocurrency Market

One of the main factors that contribute to the high price swings in cryptocurrency is the speculative nature of the asset. A speculative bubble describes a situation when the market price of an asset is different from its fundamental value because investors anticipate selling it higher for some future short-term gains, ignoring the actual returns of that asset (Iraola & Santos, 2016). Studies outline the existence of a speculative bubble in the crypto market. For example, as described by Geuder et al. (2019) the absence of actual intrinsic value and promised future payments, also combined with a lack of legal and official means of exchange characteristics, this market creates the perfect environment for speculative behaviour experiments. Yermack (2015), believes that missing properties of a unit of account and store of value are mainly due to Bitcoin's high volatility, making it a very risky asset to hold.

While discussing the volatility of the asset price, it is crucial to also examine the Efficient Market Hypothesis (EMH) proposed by Fama (1970), since asset prices reflect all available information, implying that price movements should only occur in response to new data (Fidan & Güz, 2022). However, studies point out that cryptocurrencies often differ from the EMH, revealing market inefficiencies that can explain high speculative activities within the market (Urquhart, 2016; Zhang et al., 2018).

On the other hand, the behavioural finance approach believes that investment decisions are based on “cognitive bias” (Zindel et al., 2014). According to Grobys and Junttila (2021), the lottery-like demand of taking high-risk and high-reward opportunities can be considered a key driver of speculation and volatility in the stock market is also persistent in cryptocurrency trading. The main reason for such type of demand on crypto can be described by two main inherent characteristics of this market, mentioned by Baur and Dimpfl (2018): 1) “FOMO” (Fear of missing out) trading, where inexperienced and casual traders rush to invest based on emotional reaction of missing upward trend for short term profits with no proper investment analysis. 2) “Pump and dump” schemas where a certain group artificially increases the price of the asset to sell their holdings at a high volume, which makes the price go down sharply. Both of these traits we can consider as highly speculative behaviour since these types of actions mostly rely on manipulating market trends and emotions rather than the asset's intrinsic value. This was described by Almeida and Gonçalves (2023), who mentioned that the main decision of crypto market investors is based on market sentiment. Also, the theory of liquidity suggests that assets with lower liquidity experience more pronounced price movements (Amihud & Mendelson, 1986), specifically on that Tang and Wang (2022) show that liquidity shocks are interrelated with future Bitcoin volatility, which particularly affects long-term trend components.

In general, the price history and movement of cryptocurrency are complicated to explain to a sufficient degree by standard economic practices, for the reason that the crypto market has its unique demand, which is primarily influenced by the expected gains from holding it and selling it at a higher price in the future and fixed or algorithm based supply which drives the price dynamics and partially detached from the economic and financial fundamentals, but still, there are factors of macro-financial development and macroeconomic news which attributes to the cryptocurrency price (Kristoufek, 2013; Yin et al., 2021).

1.2.2. Impact on Traditional Financial Markets

The general link and interdependence of financial asset classes can be explained by Markowitz's Modern Portfolio Theory (1952), which underlined the benefits of diversification, where the inclusion of assets with low or negative correlations in a portfolio can reduce overall risk. However, under stress conditions, financial markets may experience contagion, whereby shocks propagate throughout the economy (Harkmann, 2021), which undermines the intended benefits of diversification. This integration enables shocks that start in one market to spread to another,

especially when assets are connected through global trade and investor behaviour (Jithendranathan, 2013). The financialization of cryptocurrencies has intensified their interconnectedness with traditional markets. With increased institutional adoption, cryptocurrencies are no longer isolated from broader financial systems, instead, integration through both direct and indirect channels increases the likelihood of volatility spillovers, where shocks originating in the cryptocurrency market can propagate into other financial markets (Auer et al., 2023)

From a behavioural finance perspective, interlinkages between cryptocurrencies and other financial assets are further influenced by emotional behaviour and panic selling, particularly during market downturns. These effects might be amplified in the case of crypto markets due to many factors of speculative nature and bubbles. It could serve as a cause of the financial contagion effect where a financial crisis or significant event in one market can spread to others, leading to widespread economic disruptions (Moser, 2003). We may consider the definition more appropriately, in a cryptocurrency market sense, as an "emotional contagion effect" since negative shock could probably trigger a panic in investor behaviour and further drive significant volume movement into traditional finances for safety (Zhang et al., 2024). This kind of phenomenon is described as a "flight-for-safety" (FTS), which is a situation wherein investors, in times of risk, transfer themselves out of risky assets to safer and more liquid assets (Baele et al., 2020).

In this context, understanding how traditional financial markets—such as equities and commodities—respond to this volatility effect is very important. Since both are considered to be critical indicators of a country’s economic health. The stock market reflects the performance of its corporate sector and is closely related to the overall economic performance of the market (Levine and Zervos, 1996) and commodities markets are closely tied with modern industrialization, technological advancement and innovation (Baffes, 2022). Wang (2023) outlines that investor sentiment has a crucial role in shaping financial markets performance and investor sentiment provides an important channel through which cryptocurrency volatility can have an impact. High volatility in the cryptocurrency market can lead to changes in capital flow and have the potential to act as an effective hedge against risks compared to traditional instruments like gold or diversified portfolios (Cheong, 2019).

Furthermore, spillover effects are further complicated by the various levels of cryptocurrency adoption. Such dynamics in adoption influence the extent to which cryptocurrencies are integrated

with traditional financial systems and the potential to transmit volatility. According to Hai et al. (2023), the main factors for different cryptocurrency integrations in the country are related to the magnitude of the population and the level of financial institution development. Higher innovation, economic development, and financial inclusion are associated with greater adoption of cryptocurrencies. (Auer et al., 2023). Therefore, a deeper understanding of all underlying factors is essential for accurately assessing the spillover effects of crypto.

1.3. Empirical study evidence

Earlier studies on cryptocurrency mostly focused on the properties of price volatility and safety bubble existence within the crypto market but with the growth of the interest towards cryptocurrency, research focus expanded with the goal of understanding the spillover effects of cryptocurrency and other traditional financial markets, which challenged earlier perceptions of cryptocurrencies as independent assets.

One set of studies focused on the properties of cryptocurrency diversification, safe haven and hedge properties. The earliest studies start with Bouri et al. (2017), who used the DCC-GARCH model estimation for crypto and different traditional financial assets and found that Bitcoin can be an effective diversifier for different asset classes, but the capability of Bitcoin to hedge against traditional financial assets is limited to certain economic conditions.

Expanding on the idea from Bouri et al. (2017), Stensås et al. (2019) included one more layer of analysis by investigating developing and developed markets separately. The results from this study show that Bitcoin acts as a hedge for developing markets and a diversifier for developed markets. This study also noted a more pronounced safe haven role of Bitcoin during specific global uncertainty phases. Given the nature of cryptocurrency with amplified properties during market downturns, Tarchella et al. (2024) compared Bitcoin and Ethereum with traditional assets for the sample period of the COVID-19 pandemic, the study found that while during normal market conditions, commodities are the best hedge for equity assets, for abnormal market negative shocks cryptocurrency, show better properties for hedging. Similarly, Zha et al. (2023) concluded that there are indeed significant risk spillovers from Bitcoin to traditional financial markets during the COVID-19 pandemic outbreak.

Other strands of studies investigate spillover effects over time between crypto and the equity market. An earlier study by Trabelsi (2018) concludes that cryptocurrencies show no significant

spillover effects, and this market operates independently of traditional financial markets. On the contrary, more recent evidence by Aydoğan et al. (2024) presents evidence of a unidirectional spillover from cryptocurrency markets to the stock markets in E7 countries, while it is bidirectional in the case of G7 countries, which shows that the intensity of spillover is higher in developed markets. Expanding on the regional scope, in the analysis of African stock markets Joseph et al. (2024) point out that correlations vary across regions and levels of market development, with positive shocks tending to be more contagious. This is particularly related to the crypto adoption level within the country, since as stated in Chapter 1.2.2., economic development is one the main factors that promote cryptomarket integrity.

On the broader financial implication level, according to Umar et al. (2021), initial shocks from the cryptocurrency markets have the potential to affect all financial markets, but only specific sectors, such as equity and high-yield bond markets, exhibit persistent responses to subsequent volatility. This highlights that while the crypto market's initial volatility can have an impact across various financial sectors, its long-term influence might be restricted to specific asset classes. This selective persistence in spillover effects suggests that the interconnectedness of cryptocurrency markets with traditional financial markets is complex and not uniformly distributed (Yu & Cifuentes-Faura, 2021).

2. DATA AND METHODOLOGY

2.1. Data

Most of the studies in this specific area are based on individual cryptocurrencies' historical daily prices, but this approach can have some limitations in the interpretation of results which results in difficulties in drawing general conclusions about the “cryptocurrency market” as a whole (Zięba et al., 2019). For that particular reason, this work aims to use the daily price of the Bitwise 10 Crypto Index (ticker: BITW) as a general proxy for the cryptomarket. This will help the model to smooth out some individual noise of cryptocurrencies, due to shared market sentiments (Akyildirim et al., 2024).

In addition to the BITW index, the daily price of stock indices for six selected countries: India, Brazil, the United Kingdom, Germany, South Africa, and Japan are representative measures of traditional financial markets. These countries are chosen based on diversity in both economic structure and cryptocurrency adoption levels, allowing for a comparative analysis that is both geographically and economically representative. Table 1 details the selected split of countries, covering data from December 2, 2020, to November 8, 2024. The timespan was based on the available price history for the BITW index.

Table 1. Country split based on crypto adoption level

Level	Country	GCAI rank	Stock Market Index
High Adoption	India	1	Nifty 50
	Brazil	10	Bovespa
	UK	12	FTSE 100
Low Adoption	Germany	21	DAX
	South Africa	30	FTSE_JSE
	Japan	23	Nikkei 225

Source: Chainalysis (2024)

This comparative approach will help illustrate how different levels of market integration with cryptocurrencies impact volatility spillovers into traditional financial systems.
The split of countries was based on the Global Crypto Adoption Index (GCAI) from Chainalysis (2024) and the data for BITW, stock indices and commodities prices is derived from investing.com.

The data first were imported into R as a CSV format and later used a compiled function, which applied several steps of transformations. After that, the data is arranged chronologically by date to prepare for calculating subsequent log returns. The formula below shows the calculation for log returns:

$$log(return)_t = log(P_t) - log(P_{t-1}) \quad (1)$$

where
$log(return)_t$ - log asset return at a time t
$log(P_t)$ - log of the asset price at a time t
$log(P_{t-1})$ - log of the asset price at a time t-1

To model each variable pair for DCC-GARCH estimation, all datasets were joined using SQL logic (INNER JOIN) on the Date column, matching entries in the BITW data with each stock index, gold price and oil price dates. This will ensure an equal number of data entries for each model pair. Table 2 shows an overview of descriptive statistics of variables used in the dataset.

Table 2. Descriptive Statistics

Variable	Mean	SD	Min	Max	Skewness	Kurtosis
BITW	0,000393	0,0596	-0,4019	0,5339	0,864	13,46
DAX	0,000365	0,0103	-0,0451	0,0762	-0,033	4,447
Bovespa	0,000136	0,0115	-0,0499	0,0539	-0,196	1,037
FTSE_JSE	0,000385	0,0106	-0,0363	0,0482	0,102	1,132
FTSE100	0,000058	0,0112	-0,1151	0,0867	-1,176	16,00
Nifty50	0,000625	0,0091	-0,0611	0,0463	-0,600	4,424
Nikkei225	0,000402	0,0131	-0,1323	0,0974	-0,868	13,98
WTI	0,000368	0,0212	-0,1264	0,1027	-0,459	3,534
Gold	0,000373	0,0088	-0,0355	0,0352	-0,159	1,288

Source: author's calculations, see Appendix 1

As shown in Table 2, all 13 time series show standard deviation (SD) values generally higher than their respective mean values, highlighting a degree of variability in daily returns across datasets. Most of the variables exhibit skewness and excess kurtosis values that indicate asymmetry and deviations from normality. The results show that 5 out of 6 stock indices are left skewed, with the FTSE100 and Nikkei225 standing out with a strong negative skewness value of -1.176 and -0.868, respectively, which points to extreme negative returns in distribution. As we can see, the BITW index shows moderate right skewness and high kurtosis value, a common trait in cryptocurrency markets due to high volatility and the tendency for large price swings. However, not all variables display significant skewness. For example, DAX (-0.033), and FTSE_JSE (0.102), for instance, have low skewness values suggesting near-symmetrical distributions. For most of the time series, the kurtosis value is positively higher than 3 (leptokurtic), which is in general common to financial assets return data (Yi et al., 2018; Ben-Ahmed et al., 2023)

2.2. Methodology

The type of analysis that will be used in this work is the Generalized Autoregressive Conditional Heteroscedasticity (GARCH) model. This model was provided by Bollerslev in 1986 and this model initially arrived from Autoregressive Conditional Heteroscedasticity (Nordström, 2021). GARCH models offer significant benefits in volatility forecasting and option pricing. They provide superior volatility forecasts compared to alternative models, especially for large datasets with daily frequency (Matei, 2009), also GARCH models are adapted to the typical features of financial markets return data: non-normal residuals and excess kurtosis (Jondeau & Rockinger, 2006).

The exact form of the GARCH model in this work will be the multivariate Dynamic Conditional Correlation (DCC) GARCH model. The main difference between univariate and multivariate GARCH models is that univariate models can only be applied to one asset at a time, limiting their utility in analysing different sets of portfolios. To overcome this limitation, Engle & Sheppard (2001) proposed multivariate GARCH models like the DCC-GARCH model that captures the co-movements between multiple assets.

The estimation of the multivariate DCC-GARCH model follows a two-step approach. Firstly, we estimate univariate ARMA-GARCH models for each time series. In the mean equation, we employ

an Autoregressive Moving Average (ARMA) model proposed by Box and Jenkins (1970) to capture any serial correlation present in the return series, this is an important part of multivariate DCC-GARCH model estimation since financial return series often exhibit autocorrelation (Makridakis & Hibon, 1997), which means past values or past shocks influence current returns. The DCC-GARCH model assumes that the residuals used to model volatility are white noise, which means it is uncorrelated with constant variance (Engle, 2002). But if the model would not account for serial correlation for the mean equation, residuals will still exhibit autocorrelation, which violates the white noise assumption (Wang & Akabay, 1994). This can lead to biased and inefficient parameter estimates in the GARCH model. The AR (autoregressive) component captures the relationship between current returns and their lagged values, while the MA (moving average) component models dependencies on past shocks (Ghani & Rahim, 2019). The selection of ARMA (m,p) order presented in Appendix 1 and R function auto.arima() is used from package 'forecast' to select the best ARMA model for each time series (Hyndman & Khandakar, 2008). ARMA model is given as:

$$r_{i,t} = \mu_i + \sum_{p=1}^{P_i} \phi_{i,p} r_{i,t-p} + \sum_{q=1}^{Q_i} \theta_{i,q} \epsilon_{i,t-q} + \epsilon_{i,t} \tag{2}$$

where:
$r_{i,t}$- return of asset i at time t
μ_i- constant mean return
$\phi_{i,p}$ - autoregressive (AR) coefficients
$\theta_{i,q}$- moving average (MA) coefficients
$\epsilon_{i,t}$- residual term

Next, we model the conditional variance using the GARCH(1,1). The selection of lag order for univariate GARCH models will be GARCH (1,1) since based on several studies (see for example, Hansen & Lunde, 2003; Kalyanaraman, 2014; Magnus & Fosu, 2006), GARCH(1,1) resulted in inferior to other models due to the balance between model simplicity and the ability to capture the essential dynamics of financial time series data. The formula for univariate exponential GARCH (1,1) is given as:

$$\sigma_{i,t}{}^2 = \omega_i + \alpha_i \epsilon^2{}_{i,t-1} + \beta_i \sigma^2{}_{i,t-1} \tag{3}$$

where:
$\sigma_{i,t}{}^2$ - conditional variance of asset i at time t
ω_i - constant term
α_i - captures the short-term persistence of shocks (ARCH effects)
β_i - captures the long-term persistence of volatility (GARCH effects)

After estimating the univariate models, standardized residuals are obtained:

$$z_{i,t} = \frac{\epsilon_{i,t}}{\sigma_{i,t}}, \tag{4}$$

where:
$z_{i,t}$ is assumed to be i.i.d with zero mean and unit variance (white noise)

In the second stage, we estimate the dynamic correlations. We construct the vector of standardized residuals:

$$z_t = [z_{1,t}, z_{2,t}, \ldots, z_{n,t}]' \tag{5}$$

where:
n is the number of assets

The dynamic correlations are modelled using the following equations:

$$\mathbf{Q}_t = (1 - a - b)\bar{\mathbf{Q}} + a\mathbf{z}_{t-1}\mathbf{z}'_{t-1} + b\mathbf{Q}_{t-1}, \tag{6}$$

where:
Q_t- = time-varying covariance matrix of standardized residuals
$\underline{Q}$ - unconditional covariance matrix of z_t
α, β - scalar parameters

Further, the last step for the model will be construction of the time-varying correlation matrix:

$$\mathbf{R}_t = (\mathbf{Q}_t^*)^{-1}\,\mathbf{Q}_t\,(\mathbf{Q}_t^*)^{-1}, \tag{7}$$

where:
$Q^*{}_t$ - diagonal matrix of square roots of the diagonal elements of Qt
R_t - the time-varying correlation matrix, with all elements bounded by [-1,1]

According to Orskaug (2009), two important conditions should be met: "R_t has to be positive definite and all the elements in the correlation matrix R_t have to be equal to or less than one by definition".

Before estimating the model which is explained in Chapter 2.3, Singh et al. (2022) mention the main conditions for proper DCC-GARCH model estimation: 1) First, the dataset should be stationary. 2) Secondly, the data should have a significant ARCH effect. The ARCH effect is described by Engle (1982) as the phenomenon where the variance of a time series is conditional on past observations.

To ensure the stationary of time series, this work used the Augmented Dickey-Fuller (ADF) test proposed by Dickey and Fuller (1981) which is considered a widely used and preferable method for assessing stationarity in time series (Campbell and Perron, 1991). ADF test is run in R by using the function adf.test(). The results show that all-time series are stationary see Appendix 2. According to Sjölander (2011), the standard approach for ARCH effect presence is the ARCH-Lagrange Multiplier test proposed by Engle (1982). The test is applied on residuals after the estimation of the mean equation and to keep the results parsimonious, lag order 4 is used for the test (Brooks, 2008). According to the results of Appendix 3, all data sets at different significance levels (only for Bovespa the significance is at 0.1) reject the null hypothesis of no ARCH effect. Therefore the author can assume that the data is selected properly for GARCH model estimation. Also, the DCC-GARCH model estimates are based on the student's t-distribution, since according to Nyssanov (2013), the model estimation for univariate GARCH models with Student's t-distribution assumption outperforms those with normal distribution assumption.

The final step in the DCC-GARCH model analysis involves conducting model diagnostics to ensure its robustness. Specifically, the Ljung-Box test proposed by Ljung (1978) is applied to the standardized residuals to check for autocorrelation, and the ARCH-LM test is used on the squared standardized residuals to detect any remaining ARCH effects. These tests help confirm whether the model has successfully captured the dynamics of both the mean and volatility processes. According to Ljung (1986), the optimal lag order for the Ljung-Box test is 5 and for the ARCH-LM test lag order will follow the same approach described previously in this chapter.

2.3. Correlation matrix

Before applying the DCC-GARCH model and estimating dynamic conditional correlation, this thesis follows a similar preliminary analysis of the correlation matrix from such studies as: Ben-Ahmed et al. (2023), and Joseph et al. (2024). Figure 3 presents the results of the Pearson correlation matrix between all asset indexes used in this study.

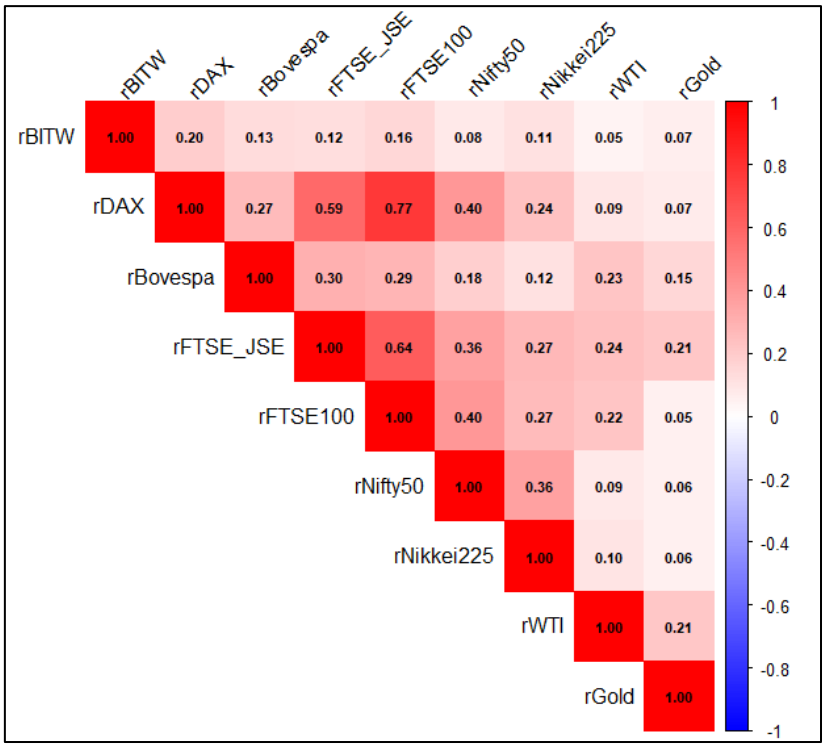

Figure 2. Correlation matrix between assets
Source: Author's calculations, see Appendix 1

The correlation matrix results are similar to those according to Joseph et al. (2024) who also found a very low correlation of cryptocurrency with stock indices in the 5 largest African countries. From Figure 2, it is visible that there is a low correlation pattern between BITW and stock indices and this suggests that cryptocurrencies and the stock market generally operate independently under normal market conditions, which is consistent with the evidence from Vo and Xu (2017), who also found a low correlation between Bitcoin and Dow Jones Industrial Average (DJIA) describing crypto as perfect risk diversifier against global financial crisis. Also, it is important to note that there is a divergence in correlations between BITW and emerging markets like Bovespa (0.13) and Nifty50 (0.08) versus developed markets like DAX (0.2) and FTSE100 (0.16), which gives preliminary insights into regional differences in market integration, aligning with Aydoğan et al. (2024), who noted stronger spillovers in developed economies. Additionally, weak correlations with commodities like gold and oil can show us some hedging aspects of crypto, which was also described by Bouri et al. (2017) who observed Bitcoin's hedging role for commodity index. However, as research studies show there is a strong integration of crypto and stock markets during market downturns and the correlation results can not provide strong evidence for possible spillover effects and dynamic correlation change during periods of stress. This particularly underscores the need for dynamic models to capture the evolving nature of relationships and volatility clustering.

3. VOLATILITY SPILLOVER ANALYSIS

This chapter presents the correlation analysis and DCC-GARCH estimation results for each asset pair, firstly focusing on two groups: low and high cryptocurrency adoption countries and later on commodities markets. Based on the methodology explained in Chapter 2, we describe the dynamic correlations between cryptocurrencies and traditional financial markets during the sample period.

3.1. DCC-GARCH model estimation

Table 3 presents the results from multivariate DCC-GARCH (1, 1) are presented. Statistical significance of values marked as: *** (0.01), **(0.05), *(0.1). Standard errors are given in parentheses.

Table 3 DCC-GARCH model estimates

Parameter	DAX	Bovespa	FTSE_JSE	FTSE100	Nifty50	Nikkei225	WTI	Gold
[A].α1	0.209*** (0.054)	0.209*** (0.055)	0.209*** (0.054)	0.209*** (0.055)	0.209*** (0.054)	0.209*** (0.054)	0.209*** (0.054)	0.209*** (0.055)
[A].β1	0.706*** (0.074)	0.706*** (0.075)	0.706*** (0.074)	0.706*** (0.074)	0.706*** (0.074)	0.706*** (0.074)	0.706*** (0.074)	0.706*** (0.074)
[B].α1	0.055*** (0.006)	0.027*** (0.001)	0.015*** (0.001)	0.187*** (0.030)	0.100** (0.043)	0.153* (0.081)	0.113*** (0.039)	0.032 (0.087)
[B].β1	0.938*** (0.009)	0.971*** (0.002)	0.980*** (0.001)	0.630*** (0.059)	0.861*** (0.075)	0.645** (0.281)	0.842*** (0.059)	0.949*** (0.103)
[Joint] dccα1	0.012** (0.005)	0.042 (0.119)	0.029 (0.039)	0.008 (0.006)	0.008 (0.006)	0.014 (0.009)	0.000 (0.000)	0.009 (0.008)
[Joint] dccβ1	0.983*** (0.006)	0.000 (1.578)	0.684 (0.558)	0.983*** (0.008)	0.976*** (0.013)	0.962*** (0.030)	0.934*** (0.158)	0.986*** (0.021)

Source: Author's calculations, see Appendix 1

Estimators denoted with [A] (BITW) and [B] (Stock indices, WTI, Gold) represent the estimations from univariate DCC-GARCH models, which are further combined as [Joint] estimators. Standard

errors are presented in brackets. Also, it is important to note that scalars : [A] < 1, [B] < 1 and [Joint] < 1, which fulfil one of the main criteria for the DCC-GARCH model (Ubaid et al., 2024)

From the univariate BITW ([A] estimator) the statistically significant ARCH term (α_1) is visible, which indicates a notable short-term reaction to shocks. The GARCH term (β_1) or long-term volatility persistence is high at 0.706 level, reflecting that the cryptocurrency market poses prolonged volatility effects.

For stock indices ([B] estimators), the results show varying sensitivity to shocks, with values ranging from 0.015 (FTSE_JSE) to 0.187 (FTSE100), all statistically significant at 1%. These values reflect the differing short-term volatility dynamics in global markets. The GARCH term (β_1) shows higher persistence, ranging from 0.630 (FTSE100) to 0.980 (FTSE_JSE), reflecting high volatility persistence, especially for DAX and Bovespa indices. For [Joint] estimators, the results show minimal immediate correlation effects, as the short-term correlation parameter (α_1) is generally insignificant, except for Germany (DAX). However, the long-term correlation persistence (β_1) is robust, with high values indicating strong long-term integration between the cryptocurrency market and traditional stock indices.

Results of the model with commodities, Gold shows statistically not significant short-term sensitivity (α_1: 0.032) and higher long-term volatility persistence (β_1: 0.949), underscoring its role as a stable, safe-haven asset. Gold's stability and low responsiveness to short-term shocks are consistent with its historical function as a hedge against market volatility and inflation. For WTI, the results indicate moderate short-term sensitivity (α_1: 0.113) and high long-term volatility persistence (β_1: 0.842). This dynamic suggests that oil prices respond more immediately to market shocks, reflecting the commodity's sensitivity to economic factors. The joint parameters highlight limited short-term spillover effects, as shown by statistically insignificant dccα_1 values, but strong long-term correlation persistence, of dccβ_1: 0.986 for Gold and dccβ_1: 0.934 for WTI, which is similar to the results from stock indices estimates.

Further model diagnostics using the Ljung-Box test and ARCH-LM test confirm the adequacy of most models. Ljung-Box test results indicate no significant serial correlation in the standardized residuals across all models, which means that model estimation properly captured mean dynamics, see Appendix 4

Additionally, most models have observed no significant ARCH effects in the squared residuals, indicating that the volatility dynamics have been properly captured. However, the ARCH-LM test for Japan (Nikkei225) and India (Nifty50) reveals significant ARCH effects ($p<0.01$), suggesting that the current model specification does not fully explain the volatility clustering in these markets. Full diagnostic results are provided in Appendix 5.

3.2. Dynamic Conditional Correlations

3.2.1. Stock markets analysis

Figure 3 represents the dynamic conditional correlations between stock indices and the BITW index from December 2021 to November 2024 for high cryptocurrency adoption countries and it shows some interesting patterns of how spillover effects changed throughout this period. For Nifty50 (India) and FTSE100 (UK), it is visible that both countries follow very similar patterns. Both indices have very weak correlations for the period of 2021 to 2022 and with further strengthening of correlations throughout 2022. In both cases graphs exhibit a steady rise, peeking around mid-2022 for Nifty50 and mid-2022 and late 2022 for FTSE100. Post-2022, the correlations for Nifty50 and FTSE100 display a downward trend, indicating a stabilization in the crypto markets and a reduction in spillover intensity, although FTSE100 show a sudden increase in correlation somewhere around mid-2023. In 2024, the correlations for Nifty50 and FTSE100 remained stable during the first half of the year, but around mid-2024, Nifty50 experienced a sharp decline in correlation, while FTSE100 showed only a slight drop, followed by an upward trend in both indices.

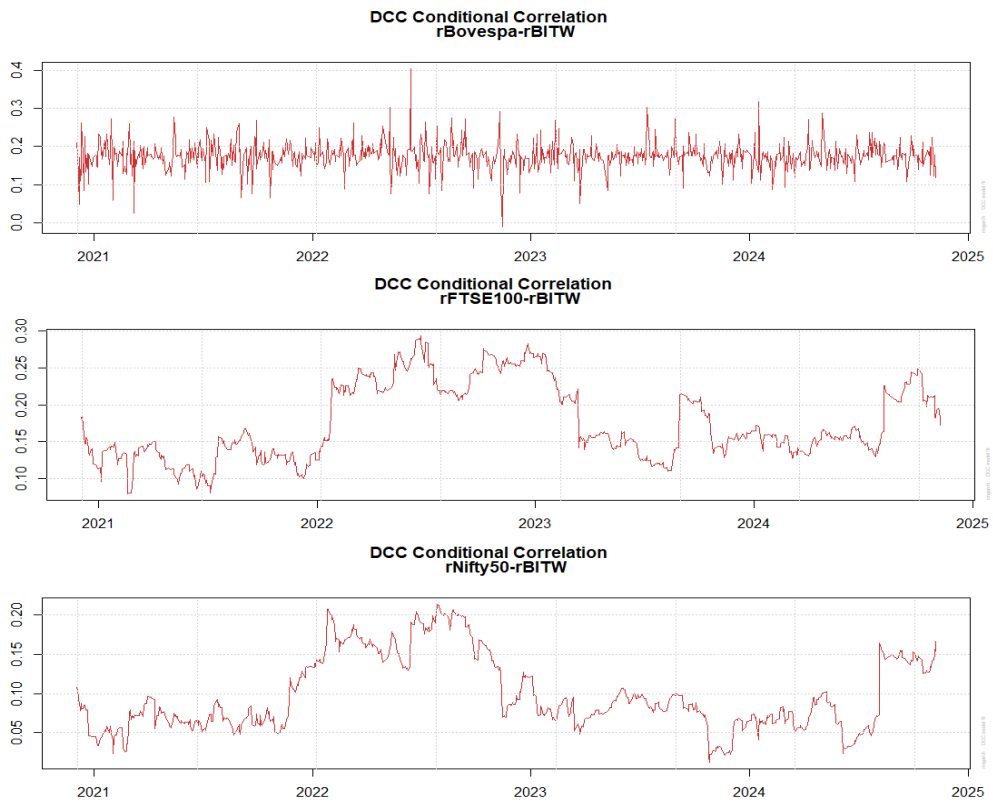

Figure 3. Dynamic conditional correlations in high adoption countries
Source: Author's calculations, see Appendix 1

Figure 4 shows the results for low crypto adoption countries and their dynamic conditional correlations over time. DAX (Germany) shows a clear upward trend in correlation with the BITW index from late 2021 through 2022, peaking close to 0.4 in early 2023. For Nikkei225 (Japan), a gradual increase in correlation is observed from 2021, peaking somewhere during mid-2023 at around 0.25, with a later sharp decline. Nevertheless, Nikkei225 follows a similar pattern with high crypto adoption countries with a sharp increase in mid-2024, while DAX on the other hand showed a more stable constant increase for the correlations observed in 2024.

As seen by graphs, India, the UK, Japan and Germany show very similar correlation dynamics for the sample period, which is interestingly coincident with some combination of global and local events in the crypto market. First of all, it is important to mention that through 2021 Bitcoin had significant growth and in November 2021 reached an all-time high peaking at 69000 USD (Kollewe, 2021) which is the moment when the correlation pattern started to increase, and might be one of the starting mechanisms that pushed crypto and stock market increased integration. Later on, the year 2022 was rich with some major global and local events that affected the crypto market and possibly created a more pronounced contagion effect on stock markets.

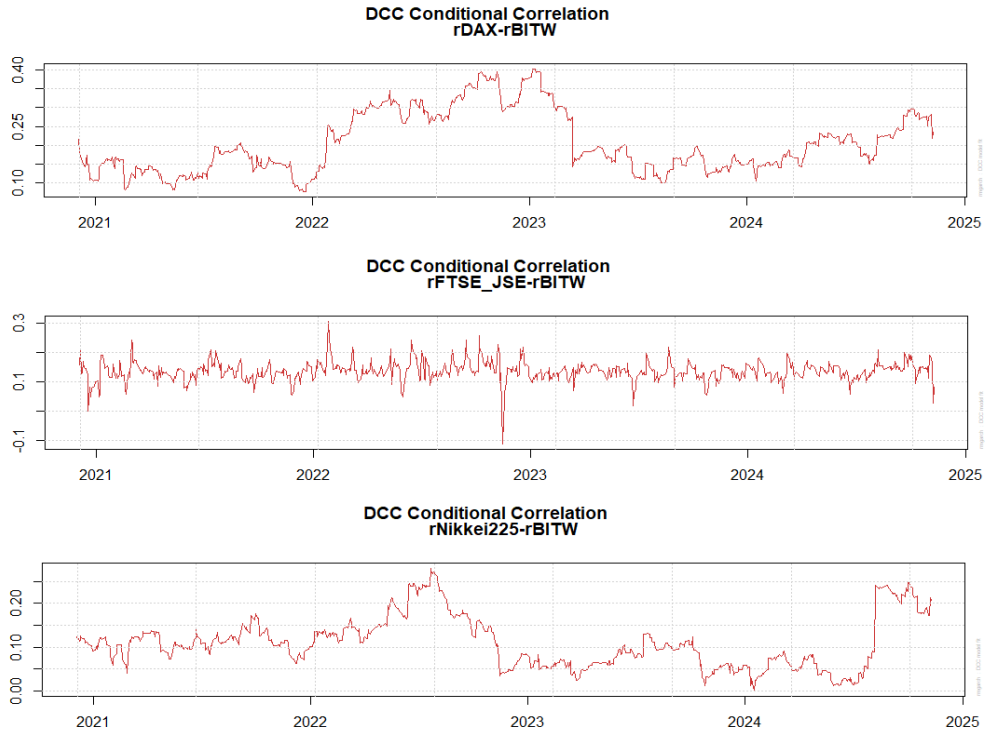

Figure 4. Dynamic conditional correlations in low adoption countries
Source: Author's calculations, see Appendix 1

Global events that mainly interrupted cryptocurrency market prices are the FTX collapse in November 2022 (Conlon et al., 2022), which caused a dramatic revaluation of risk assets globally and the Terra (LUNA) collapse in May 2022 (Liu et al., 2023) added another layer of uncertainty, leading to sharp market declines and investor flight from high-risk assets. These global shocks likely amplified the upward trend in correlation during 2022, as seen in the peaks for stock indices during that time reflecting a synchronized movement between crypto and stock markets in response to the crisis events, and also according to Aibai et al. (2024), spillover effects from bad volatility in general exceeds those from good volatility. After 2022, the pattern switched to a downward trend, this is particularly related to the "roller-coaster" events from 2022 which led to crypto market stabilization in 2023 (Borisova, 2023). In 2024, as mentioned by CryptoGlobe (2024), investors are more interested in crypto as a growing and more recognized asset class and this interest is driven by the combination of some events such as the US Bitcoin ETF trading announcement, Bitcoin record high value and halving of Bitcoin.

From local events it is worth noting some structural and regulatory changes in those markets, for example in April 2022 UK announced the goal to become a "global hub" for crypto assets (Dewey, 2024), meanwhile in India announced some policy changes such as India's crypto taxation

framework in early 2022 (Arora & Kaur, 2024), and in Germany policy update for the sale of purchased bitcoin and ether tax-free after one year for private individuals (Hicks, 2023). Japan had a very distinct sharp increase in volatility, which is not observable in other countries, this can be related to the event when DMM Bitcoin was hacked and approximately 48.2 billion JPY worth of Bitcoin was stolen (Reuters, 2024). But one important note is that these events to some extent may contribute to correlation dynamics, but cannot be viewed in isolation as primary drivers of the observed spillover effects. Instead, they form part of a complex picture where multiple factors collectively shape how markets move to one another.

For Bovespa (Brazil) and FTSE/JSE (South Africa), the author found no consistent spillover pattern for the sample period. Only somewhere close to late 2022 Bovespa index correlation reach an episodic spike of ~0.4, while somewhere around Q4 in 2022 FTSE/JSE dropped to a negative correlation level. The values fluctuate frequently, without showing a clear long-term trend. That author suggests that the relationship between the South African and Brazilian stock markets and the cryptocurrency market is more episodic and less structured compared to other countries, which makes sense since spillover effects of crypto might be more pronounced in developed regions rather than developing markets (Aydogan et al., 2024).

3.2.2. Commodities market analysis

Dynamic conditional correlations for commodities markets are presented in Figure 5. It shows that, within the period 2021-2024, there was almost no correlation between WTI crude oil and Bitcoin (BITW), as dynamic correlation values were below ~0.064. This result contradicts the conclusions of Zha et al. (2023), who noticed that during the pandemic period, significant spillover dynamics between Bitcoin and crude oil were detected and further risk spillover enhancement after the pandemic. But so far we have some evidence to confirm previous studies, such as Chen and Ong (2023), who outlined that Bitcoin has a weak linkage with commodity markets and this should be considered for diversification properties and Lu et al. (2022) who found that interconnectedness between crypto and oil increased significantly during COVID-19 outbreak and post COVID-19 become even lower compared to sample starting period.

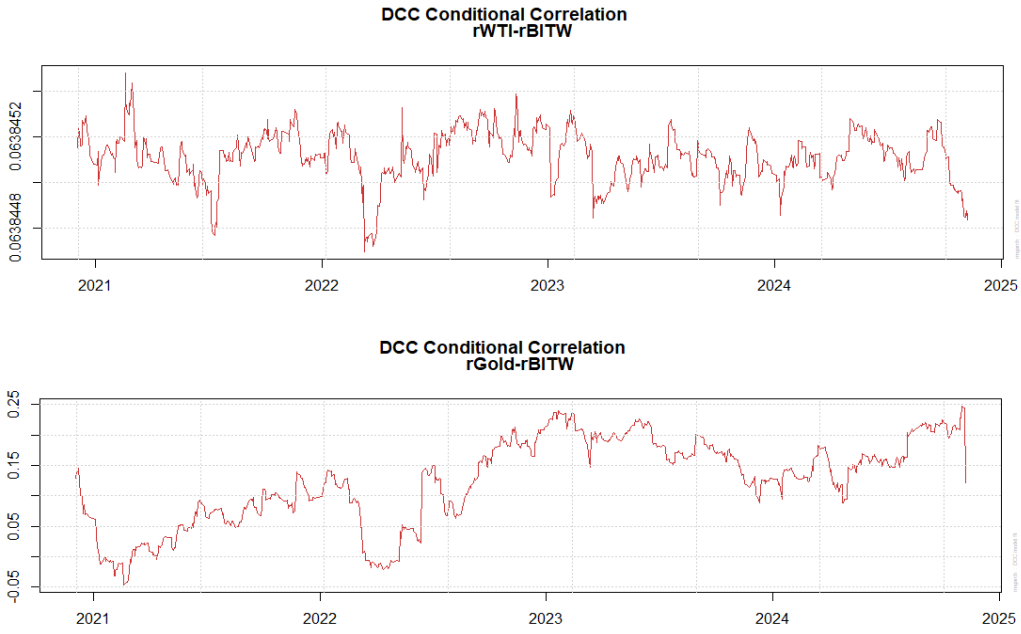

Figure 5. Dynamic conditional correlations for WTI and Gold
Source: Author's calculations, see Appendix 1

The correlation of Bitcoin and gold in early 2021 and 2022 was slightly negative, at about -0.05, this is a situation that supports the traditional role of gold as a hedge against volatility. However, from the middle of 2022, it gradually rose, reaching its peak at about 0.25 in 2023 and maintaining a stable range from 0.2 to 0.1. On similar results, Dyhrberg (2016), outlined that Bitcoin shares common hedging characteristics as gold, with a near identical reaction to good and bad news and Zha et al. (2023) mentions that after the pandemic gold and Bitcoin correlation dynamics was always positive and more volatile.

While analysing positive and negative shock effects on correlation dynamics, it is important to consider that the estimated model for standard GARCH ignores asymmetric relationships between assets (Ampountolas, 2023). For that reason, the author additionally employed Engle and Ng's (1993) sign bias test, which is used to capture asymmetric leverage within univariate GARCH models (Donaldson and Kamstra, 1997). From the test results, we can observe that there is enough evidence of a negative leverage effect, with significant Negative Sign Bias test results for Nikkei225, WTI, Bovespa and FTSE_JSE (see Appendix 6).

3.3. Discussion

The results indicate that when comparing countries with high and low levels of cryptocurrency adoption, there are no clear or consistent dynamics. This challenges the initial assumption of this study that the level of adoption of cryptocurrencies significantly affects the degree of volatility transfer between markets.

For both markets, DCC-GARCH estimators showed that short-term spillovers were largely insignificant across markets, with only the DAX index being an exception and showing a statistically significant short-term effect. In general, the results indicate that the cryptocurrency market does not exert immediate volatility transmission to most traditional financial markets. However, long-term volatility was significant for most asset classes. This emphasises that the long-term impact of cryptocurrency volatility on these markets is more pronounced than its immediate effects. In general, these results are consistent with prior studies, which outline that the short-term effects of crypto on stock markets can be limited, but can manifest in the long run (Bhatia and Bedi, 2022; Singh et al., 2022; Mishra and Dash, 2024; Ibrahim et al., 2024). This limited direct transmission of volatility from cryptocurrencies to the traditional market in the short term can be explained by the relative independence of these markets under normal conditions (Trabelsi, 2018; Mensi et al., 2023).

The dynamic conditional correlation over time, presented in Chapter 3.3, reveals that there is no specific distinct pattern for spillover effects from crypto to stock in low and high-crypto adoption countries. With the results of stock indices, the high-adoption countries and low-adoption countries, in general, followed a similar pattern of movement. For India, the UK, Germany and Japan. Correlations increased during 2022, likely due to significant global market events like the Terra (LUNA) and FTX collapses and some regional police changes and further decline of correlation with market stabilization in 2023, later on with an increase in 2024. These results are mainly consistent with previous studies (Bouri et al., 2017; Stensås et al., 2019; Tarchella et al., 2024; Joseph et al., 2024) which found that the correlation between Bitcoin and global stock markets increases during periods of financial turmoil, indicating a time-varying relationship influenced by market stress.

But, even though this study found that the periods of correlation increase tend to follow the events from cryptocurrency, it is important to note that the original assumption for low and high crypto

adoption split and the spillover effects to stock markets is very simplified, and considers that the stock market movement is driven by local investors actions, this might not necessarily be the case. While it can be logical to assume that countries with higher crypto adoption would show more sensitive responses to crypto shocks—perhaps due to home bias, (Wallmeier and Iseli, 2022), studies show that this is not exactly the case for India and UK (Oehler et al., 2017; Mukherjee et al., 2018). As a result, correlation shifts can occur regardless of a country's crypto adoption level and global factors and cross-border investment flows likely play a more significant role

The commodities market results while BITW shows some evolving relationships with traditional safe-haven assets like gold, but its spillover effects on oil markets remain limited and context-dependent. These findings suggest that cryptocurrency's role in the broader financial ecosystem is still developing and varies across asset classes (Yu & Cifuentes-Faura, 2021; Umar et al., 2021)

Several factors limit the results of this work. First, the reliance on the BITW index, while beneficial for capturing market-wide trends, may smooth out individual cryptocurrency dynamics. Secondly, expanding the analysis to include more countries with varying cryptocurrency adoption levels and significant home bias levels could reveal more insights into regional dynamics. Thirdly, the results of the Engle-Ng test show that some of the univariate models failed to capture asymmetry in the data for some models, therefore incorporating models that account for asymmetry, would better capture the complexities of volatility spillovers. Additionally, this work was based on daily closing prices of each time series, and this created a problem of asynchronous data between datasets due to different trading hours.

CONCLUSION

The original purpose of this work was to understand 1) whether cryptocurrency market volatility, has a significant impact on stock markets in countries with high and low cryptocurrency adoption, and 2) whether cryptocurrency volatility affects commodity markets, in particular gold and WTI crude oil prices, using the return data for BITW index, gold, WTI crude oil and stock indices. By estimating the DCC-GARCH model, the analyses showed that while there were notable long-term spillover effects, the short-term impacts tended to be small and insignificant. Furthermore, the analysis comparing countries with high and low crypto adoption revealed no clear or consistent differences, since

Dynamic conditional correlations showed that crypto and financial markets are largely independent under normal market conditions but become more integrated during negative or positive market shocks. The commodity market showed limited spillover effects, with cryptocurrency showing a stronger relationship with gold compared to oil.

An important finding of this study is that cryptocurrencies and traditional financial markets are independent under normal conditions, but significant market events can fuel closer integration. The study results show that in 2022 crypto assets had an increasing pattern of correlation dynamics with stock markets and gold—coinciding with a period of notable volatility within crypto markets. Nonetheless, the analysis of dynamic conditional correlation patterns did not reveal any significant differences between high- and low-adoption countries. This suggests that global capital flows are more critical in driving cryptocurrency contagion to traditional financial markets than regional adoption levels.

Several limitations, including the reliance on the BITW index and the exclusion of asymmetric volatility models, indicate areas for future research. Expanding the analysis and addressing all limitations in this study could provide deeper insights into the complex dynamics between cryptocurrency and traditional financial markets. Overall, this thesis contributes to the

understanding of cryptocurrency's role in financial stability and highlights the importance of considering both time-based and structural factors in volatility spillover analyses.

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APPENDICES

Appendix 1. Link to R-code and data

https://drive.google.com/drive/folders/1ST0Qyh7bZraIZnNnp8rgKrC6zZP-LGDC?usp=drive_link

Appendix 2. Robustness check: Results of ADF test

Time-series	Dickey-Fuller value	p-value
BITW	-13,861	<0.01***
DAX	-8,8885	<0.01***
Bovespa	-9,3568	<0.01***
FTSE_JSE	-11,322	<0.01***
FTSE100	-10,262	<0.01***
Nifty50	-10,652	<0.01***
Nikkei225	-10,998	<0.01***
WTI	-11,871	<0.01***
Gold	-10,284	<0.01***

Alternative hypothesis: stationary
Statistical significance of values marked as: *** (0.01), **(0.05), *(0.1).

Source: Author's calculations, see Appendix 1

Appendix 3. Robustness check: Results of ARCH-LM test before model estimation

Time-series	Chi-Squared	p-value
BITW	255,59	< 2.2e-16***
DAX	78,156	4.282e-16***
Bovespa	8,6474	0.07054*
FTSE_JSE	11,094	0.02553**
FTSE100	64,524	3.241e-13***
Nifty50	67,231	8.713e-14***
Nikkei225	243,57	< 2.2e-16***
WTI	38,980	7.034e-08***
Gold	9,8863	0.04239**

ARCH LM-test; Null hypothesis: no ARCH effects
Statistical significance of values marked as: *** (0.01), **(0.05), *(0.1).

Source: Author's calculations, see Appendix 1

Appendix 4. Robustness check: Results of Ljung-Box test

Time-series	Chi-Squared	p-value
BITW	2,1134	0,8332
DAX	2,4014	0,7913
Bovespa	3,2086	0,6679
FTSE_JSE	5,7271	0,3337
FTSE100	4,6330	0,4623
Nifty50	5,6882	0,3378
Nikkei225	6,4998	0,2606
WTI	6,0009	0,3061
Gold	3,2013	0,6690

Ljung Box-test; Null hypothesis: No significant autocorrelation
Statistical significance of values marked as: *** (0.01), **(0.05), *(0.1).

Source: author's calculations, see Appendix 1

Appendix 5. Robustness check: Results of ARCH-LM test after model estimation

Time-series	Chi-Squared	p-value
BITW	3,7585	0,4397
DAX	1,4685	0,8322
Bovespa	4,3772	0,3574
FTSE_JSE	1,6316	0,8031
FTSE100	0,3972	0,9827
Nifty50	14,580	0,0057***
Nikkei225	16,852	0,0021***
WTI	1,8879	0,7564
Gold	3,9231	0,4165

ARCH LM-test; Null hypothesis: no ARCH effects
Statistical significance of values marked as: *** (0.01), **(0.05), *(0.1).

Source: author's calculations, see Appendix 1

Appendix 6. Robustness check: Results of Engle-Ng test

Time Series	Sign Bias	Negative Sign Bias	Positive Sign Bias	Joint Effect
BITW	1.434 (0.152)	0.769 (0.442)	0.009 (0.993)	2.645 (0.450)
DAX	0.915 (0.360)	1.092 (0.275)	0.484 (0.628)	7.278 (0.064)*
Bovespa	1.956 (0.051)*	2.291 (0.022)**	1.030 (0.303)	6.396 (0.094)*
FTSE_JSE	0.473 (0.636)	1.846 (0.065)*	0.456 (0.648)	7.249 (0.064)*
FTSE100	0.369 (0.712)	0.420 (0.675)	1.614 (0.107)	3.824 (0.281)
Nifty50	0.908 (0.364)	1.215 (0.225)	1.165 (0.244)	4.423 (0.219)
Nikkei225	0.744 (0.457)	3.928 (0.000)***	0.869 (0.385)	21.310 (0.00)***
WTI	1.627 (0.104)	2.139 (0.033)**	0.035 (0.972)	5.312 (0.150)
Gold	0.383 (0.702)	1.035 (0.301)	0.654 (0.513)	2.098 (0.552)

Statistical significance of values marked as: *** (0.01), **(0.05), *(0.1). (p-value presented in parentheses).

Source: author's calculations, see Appendix 1

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